

# PUTTING ENTREPRENEURIAL THEORIES INTO ACTION: INSIGHTS FROM A MATCHED STUDY OF THREE SOLAR VENTURES

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**Research summary.** *The emerging theory-based view depicts entrepreneurs as sophisticated thinkers who form strategies based on rich causal theories. Existing research demonstrates the value of training entrepreneurs to articulate theories and use experiments prior to launch. Yet, entrepreneurship is emergent and adaptive, and we know less about how entrepreneurs translate their theories into action after launch. We address this gap through a longitudinal study of three matched ventures in the U.S. solar industry from 2007-2016. Our inducted model illuminates how theories are enacted – i.e., translated into action and then updated (or not) over time. We contribute by offering a novel empirical operationalization of theories, unpacking a surprising asymmetry in how theories are updated, and highlighting the role of theories in surfacing and justifying novel opportunities over time. (125)*

**Managerial summary.** *Entrepreneurs are often depicted as forming strategy based on rich causal theories. A growing body of research demonstrates how training entrepreneurs to hold and refine theories, often through experimentation, improves venture performance. Yet, entrepreneurship is emergent and adaptive, and we know less about whether or how entrepreneurs continue to translate their theories into action after launch. Through a study of three entrepreneurial ventures in the U.S. solar industry over ten years (2007-2016), we examine this process. Our model contributes to research and practice by illuminating a surprising juxtaposition, with entrepreneurs updating only selective elements of their theories, and by uncovering how theories – even when resistant to change – may nonetheless enable entrepreneurs to surface and justify novel opportunities over time. (120)*

**Keywords:** Entrepreneurship, theory-based view, strategy formation, solar

## INTRODUCTION

An emerging body of work on the theory-based view of the firm holds that entrepreneurs trying to identify novel opportunities, and effective strategies to capture them, benefit from developing “theories” (Felin & Zenger, 2009). Theories are generally understood to be mental models of business ideas; they articulate a potential future and the various assumptions, contingencies, or actions that might allow that future to be realized (Camuffo et al., 2024; Ehrig & Schmidt, 2022; Felin & Zenger, 2017; Wuebker et al., 2023). Theories facilitate entrepreneurial strategy formation by sensitizing entrepreneurs to novel opportunities or gaps in the market (Felin & Zenger, 2016; Zellweger & Zenger, 2023). They also help structure learning, in particular by providing a cognitive framework for conducting what-if analysis and counterfactual reasoning, identifying hypotheses, and interpreting the results of experiments (Agarwal et al., 2025; Ehrig & Schmidt, 2022; Wuebker et al., 2023). In this sense, entrepreneurial theories play a parallel role to scientific theories – with scholars often explicitly using the term “scientific entrepreneurship” (Camuffo et al., 2020; Zellweger & Zenger, 2023).

Literature on the theory-based view falls into two primary streams. The first is largely conceptual, and builds out the descriptive and normative elements of the theory-based perspective (Ehrig & Schmidt, 2022; Felin & Zenger, 2009; 2017; Zellweger & Zenger, 2023). These studies provide a description of how entrepreneurs might (or should) articulate theories, the characteristics of effective entrepreneurial theories, and the specific roles that theories might play in facilitating opportunity recognition and firm strategy. A second stream is largely empirical. Studies in this stream use randomized control trials (RCTs) to capture and estimate the precise benefits of training entrepreneurs to a) articulate guiding theories, and b) refine their theories by conducting experiments (Agarwal et al., 2025; Camuffo et al., 2020; 2024; Novelli &

Spina, 2024). This work offers substantial evidence of the value of theories and experiments, finding positive impacts on revenue (Agarwal et al., 2025; Camuffo et al., 2024; Novelli & Spina, 2024), profit (Agarwal et al., 2025), and survival (Camuffo et al., 2020; 2024).

Yet, despite these valuable insights, gaps remain. One important gap is that existing (empirical) research has not yet traced how entrepreneurial theories, once articulated, are put into action following launch and over time. We know, however, from the rich literature on entrepreneurial processes that strategy formation is emergent and adaptive (Kirtley & O'Mahony, 2023; Ott et al., 2017; Pillai et al., 2020). This is particularly true when addressing nascent markets, untested technologies, or novel business models, as entrepreneurs in these contexts are likely to face tremendous uncertainty, with little clarity into what is possible (McMullen & Shepherd, 2006; Packard et al., 2017). Moreover, even deliberate testing is unlikely to completely resolve these uncertainties prior to launch, as some learning may be possible only through actual participation in the market (Gans et al., 2019; Pillai et al., 2020), and actions taken to execute a strategy often reveal new, unexpected information (Bingham, 2009). Thus, even when entrepreneurs validate their theories prior to launch, they will likely need to adapt *following* launch and over time. Existing research has yet to explore whether or how this occurs. With this in mind, we ask: *How are entrepreneurial theories enacted (i.e., translated into action and then refined in light of new information) over time?*

Examining this question requires analysis of firm activity over a longer period of time than studies that examine experimentation at and prior to launch. We therefore conduct a multiple-case inductive study of three entrepreneurial ventures in the U.S. solar industry from 2007 to 2016 (Eisenhardt & Graebner, 2007). A multiple case study allows us to trace the theories and strategies of three firms in close detail over a long period of time, thus providing

insight into how theories and strategies evolve, even well after launch. The three ventures we trace were founded at the same time in the same market, but had founding teams that held radically different theories about what problems to solve and how to solve them. Using archival materials (personal blog posts, social media, books, and interviews) as well as primary interviews, we trace the evolution of these theories, the actions undertaken as a function of these theories, and the way in which the theories were revised (or not). In this way, we utilize the contrast between firms and over time to illuminate how theories are enacted.

We contribute by bridging research on the theory-based view and research on entrepreneurial strategy formation. Our first contribution is to develop a novel operationalization of entrepreneurial theories, grounded *in situ*. Our three-part definition (problem, logic, and solution) integrates and extends prior conceptual and empirical work on the theory-based view. It integrates prior work by foregrounding the causal *logics* as a central element underpinning entrepreneurial theories – an element that is core to conceptual work (e.g., Felin & Zenger, 2017) but often not captured in empirical work. It extends prior work by highlighting a critical distinction between *core* and *peripheral* beliefs, and by highlighting the possibility of a one-to-many relationship between problems-logic pairs and *many* potential solutions. A second contribution is to bring a more nuanced understanding of learning over time to the theory-based view. Prior work envisions theories as readily updated in light of new information. We find instead that learning is selective and asymmetric, such that theories exhibit a predictable degree of stability over time. Specifically, we see entrepreneurs revising beliefs about *how to organize* (i.e., solutions) *without* revising beliefs regarding *problems* to solve or the *logics* linking problems and solutions. Finally, we contribute by unpacking the counterintuitive implications of this stability. Whereas prior work often equates stable beliefs with strategic *inertia* (e.g., Tripsas

& Gavetti, 2000), we find that stable beliefs may allow strategic *momentum* – continued, patterned change over time. Through this finding, we offer insight into not only the persistence of heterogeneity within industries, but also sustained competitive advantage.

## BACKGROUND

Our research interest is in how theories, once articulated and refined by entrepreneurs, interact with firms' strategic actions over time. *Entrepreneurial theories* are cognitive representations or mental models of a business idea and the conditions or actions under which this idea might be realized (Ehrig & Schmidt, 2022; Felin & Zenger, 2009). Theories are valuable because they sensitize entrepreneurs to novel opportunities or gaps in the market, guide the formation of hypotheses and the interpretation of information, and inform the strategies that a venture might pursue (Agarwal et al., 2025; Wuebker et al., 2023). Consistent with prior literature, we define *strategy* as the set of activities through which firms attempt to create value and succeed in the marketplace (Ott & Eisenhardt, 2020; Porter, 1996). Thus, whereas theories reside entirely within the minds of entrepreneurs, strategies reflect the actions and resource commitments undertaken by the firm to create and capture value. Our interest is the interplay of entrepreneurial theories and real-world strategic actions.

Several streams of research provide valuable insight into this phenomenon. The first is an important set of conceptual papers on the theory-based view (e.g., Felin & Zenger, 2009; Sorensen, 2024; Zellweger & Zenger, 2023). This work establishes a descriptive and normative foundation for the role of theories by articulating how entrepreneurs form theories, the characteristics of effective theories, and the relationship between theories and firm strategy. Theories are understood to emerge from a question or problem, which in turn “*prompt[s] a novel hypothesis or conjecture about paths to a solution*” (Felin & Zenger, 2017, p. 262). Combining

experience and observation with logical deduction and imagination, entrepreneurs articulate theories as a set of beliefs about the value to be created and actions needed to realize this value (Ehrig & Schmidt, 2022; Wuebker et al., 2023). Entrepreneurs then *refine* this nascent theory using logical deduction (Ehrig & Schmidt, 2022; Sorenson, 2024), and by validating hypotheses through experiments (Felin & Zenger, 2017). As (or if) testing reveals unsupported or inconsistent beliefs, entrepreneurs update their theories, in line with a fundamentally “scientific” approach (Felin & Holweg, 2024; Zellweger & Zenger, 2023).

A second stream is the empirical work on scientific entrepreneurship. This research offers insight into the benefit of forming and refining theories during the early stages of the entrepreneurial process, using experiments (e.g., Camuffo et al., 2020; Gambardella & Messinese, 2025). Studies in this stream use randomized control trial (RCT) research designs to investigate performance differences between firms that use theories and perform experiments vs. firms that do not. For example, Camuffo and colleagues (2024) describe a set of four field experiments with 759 startups in Italy and the UK. Entrepreneurs in the treatment group were trained to articulate a theory using the Business Model Canvas, to identify underlying hypotheses, and to use experiments to validate them. Entrepreneurs in the control group were left to follow their intuition. This “scientific” training helped entrepreneurs by reducing overoptimism and encouraging them to consider a wider set of alternative scenarios, which led to faster termination, increased revenue, and more effective pivots. Drawing on a subset of the same sample, Novelli and Spina (2024) extend these findings, observing that a) treated entrepreneurs grew larger firms, and b) the benefit of treatment was greater (in size and revenue) for ventures with more advanced business models. In contrast, earlier stage entrepreneurs actually suffered from adopting a scientific approach in the short-term, as they were more likely

to go back to the drawing board to revise fundamental aspects of their ideas.

Recent research extends this work, for example by disaggregating the benefits of theorizing and experimentation. Agarwal and colleagues (2025) use an RCT design and a sample of 151 Tanzanian entrepreneurs and find that, relative to entrepreneurs who had been trained only to test their ideas, those trained to articulate a theory prior to experimentation achieved higher revenues and profits and were more likely to make coordinated changes to their business models. Valentine and colleagues (2024) study 261 entrepreneurs in the UK and find that engaging in either formal theorizing *or* formal experimentation alone can be problematic. Specifically, formal theorizing without experimentation may lead to biases and mistakes, whereas formal experimentation without theorizing can result in reactive pivots that lack a stable base. Together, however, the two practices increase the likelihood of effective pivots (Valentine, et al., 2024). Other work identifies boundary conditions or compares the scientific approach with alternative approaches. For example, Gagliardi and Novelli (2025) examine 196 female entrepreneurs and find that a scientific approach may be particularly effective for entrepreneurs building ventures within the domain of their own experience, suggesting that it helps address the “knowledge corridors” associated with prior experience (e.g., Agarwal & Shah, 2014).

Overall, research in this area a) articulates a sophisticated model within which entrepreneurs form and then refine theories using experiments, and b) finds persuasive evidence that training entrepreneurs to form and test theories improves venture performance. A key strength of these studies is their use of large samples and rigorous causal designs to isolate and estimate the effects of a scientific approach. In contrast, this research does not (and is not intended to) actually trace the process by which theories are enacted. That is, it neither examines how entrepreneurs move from their initial idea to an actual, real-world strategy and set of

resource commitments, nor whether or how theories are updated based on these actions – rather than experimentation – as the firms succeed, fail, or muddle through.

This lacuna is important given a third stream of research, which is that on entrepreneurial process and strategy formation (e.g., Bingham, 2009; Ott & Eisenhardt, 2020). Broadly, this work examines how entrepreneurs identify effective strategies, especially in settings characterized by uncertainty and ambiguity. A key finding is that while mental representations (like theories) are helpful to guide action and maintain coherence among activities, strategy formation is unavoidably emergent and adaptive (Ott et al., 2017). Particularly for new products and new markets, understanding *what is possible* may not be discernible through experimentation or observation alone. Instead, firms must often develop technologies, launch products, and compete in the market (Gans et al., 2019; Shelef et al., 2025). For example, neither the specific technologies nor genres of games that might eventually underpin the nascent wireless gaming industry were clear until entrants invested the resources to build and launch products (Ozcan & Eisenhardt, 2009). Similarly, early automakers had to design and manufacture vehicles to understand consumer preferences and identify which technologies were viable in real-world operating conditions (Pillai et al., 2020). In this view, entrepreneurs are likely to follow a “meandering course” (Hastie, 2001: 665) that involves continual learning and adjustment over time (Packard et al., 2017) – and not through experiments alone (Bingham, 2009). Indeed, Pillai and colleagues argue this point explicitly, stating that “*The learning that happens through market participation cannot be replaced by laboratory experiments, manufacturing prototypes, marketing reports, or conducting surveys of potential customers*” (Pillai et al., 2020: 373). This is especially true in nascent and emerging industries, or those characterized by rich interdependencies, which are understood to make it difficult to predict a



*priori* what might work (Gans et al., 2019; Rosenberg, 1994).

Collectively, these research streams are insightful but incomplete. Extant empirical work provides robust evidence that articulating and refining theories through experiments improves performance. Related conceptual work also offers a sophisticated model of how this learning might occur. Yet, we know little about how entrepreneurs translate their theories into strategies and strategic actions; nor do we know whether theories remain useful as entrepreneurs unavoidably encounter surprises and learn “by doing” over time. Indeed, learning by doing represents a challenge for the contrarian “*bets against the market*” that are understood to lie at the heart of effective theories (Felin & Zenger, 2017, p. 262), as novel theories – and the strategies they inform – are likely to face pushback from stakeholders (Adner & Levinthal, 2024; Zuzul & Edmondson, 2017). How then do entrepreneurs update their theories, in light of inevitable setbacks? More broadly, mental representations (like theories) are useful for guiding action, but can also blind actors to new information, especially if it is inconsistent with existing beliefs (e.g., Tripsas, 2009; Tripsas & Gavetti, 2000). Thus, although the theory-based view depicts theories as readily revised, might entrepreneurial theories, once articulated, blind entrepreneurs to emergent possibilities? And if so, how do ventures continue to adapt and pivot, as is critical in nascent settings (Kirtley & O’Mahony, 2023; Ott et al., 2017)? Overall, this points to the need to better understand how theories are put into action, how they are revised (or not), and how theories interact with strategy.

## METHODS

The objective of our study – to build and deepen theory around the *process of* how entrepreneurial theories guide strategic action (and how action guides theory) over time – warrants a qualitative approach (Eisenhardt & Graebner, 2007; Langley, 1999). Specifically, we

use a multiple-case study with a matched-case design (Bechky & O'Mahoney, 2015; Volmar & Eisenhardt, 2025), wherein we study a matched set of three ventures in the same industry and with other key similarities, but which vary on key constructs of interest (i.e., theories and strategies). Relative to a single case, a multiple-case approach allows us to better isolate mechanisms and interactions utilizing the contrast between cases, improving the precision of our findings and lessening the chance of overdetermining our theory (Eisenhardt, 2021).

### **Research setting: The U.S. solar industry**

Our research setting is the U.S. solar industry from 2007-2016. While the underlying technology is over half a century old, solar photovoltaic generation played a negligible role in the U.S. electricity sector until the mid-2000s. For example, the residential solar market grew from 30,000 *cumulative* installations in 2005 to 540,000 *annual* installations in 2015, and 92% of solar capacity (residential and commercial) was built between 2010 and 2015 (SEIA, 2016).

This explosive growth was driven by several factors relevant to our study. The first was falling panel prices, which decreased overall system costs and expanded the size of the total addressable market. A second factor was the emergence of consumer-accessible satellite photography, which enabled engineers for the first time to design and sell solar systems without visiting sites in person. This reduced costs, while also introducing the novel possibility of installation, sales, and system design activities being *unbundled* across firms (historically, these activities had been performed together). Third, there was an explosion of state and federal incentive programs and policies to support the industry. One particularly significant policy was the 2005 Energy Policy Act, which allowed the *owners* of solar systems to claim a tax credit equal to 30% of the system cost. The assignment of this tax credit to owners (versus operators or installers) opened an entirely new means of financing installations in “leased” or third-party

owned (TPO) solar, which potentially allowed a range of new business models.

The U.S. solar industry is an attractive research setting for several reasons. First, the explosive growth witnessed over the study period, paired with the raft of new entrants and novel business models that it sparked, introduced massive uncertainty in terms of which strategies, products, and even markets would be viable. In articles written over the entire study period, experts made claims like “*the solar market is anyone’s game*” and debated the merits of different products and strategies, as well as the growth potential of various submarkets (e.g., Kanellos, 2009; Shah & Litvak, 2016). More broadly, the importance of solar as a technology and industry critical to a range of issues (e.g., energy security, climate change) *and* its attractiveness in terms of rapid growth presented founders with a range of possible problems to address. This openness meant that any number of theories were viable, and that learning about which might work and how was paramount – a perspective reflected in our founder interviews as well. Solar is also an attractive research setting because it is a rich ecosystem, with five distinct components (i.e., sales, installation, finance, panels, and racking), as well as three distinct customer markets (i.e., residential, commercial, utility). Complexity limits actors’ ability to predict which markets or products will succeed, while also increasing the likelihood of disruption (Adner & Kapoor, 2010; Ozcan & Hannah, 2020), making learning and adjustment particularly likely. Finally, the industry is well-documented, with founder perspectives and analyst assessments captured by a variety of cleantech-focused blogs (e.g., Greentech Media, GigaOM) and video-sharing sites (e.g., YouTube) that were launched at the time. These resources provide rich data from which to capture the evolution of theories and strategies over time.

### **Matched case design**

We use a matched case design. This research design consists of a small set of cases which

share many similar features that allow comparability, but which also have key differences relevant to the theoretical interest (Bechky & O'Mahony, 2015). Such designs offer the benefit of being able to observe replication of mechanisms and constructs across multiple cases, while allowing for rich presentations of each case and helping “control for” some alternative explanations due to the similarity across cases (Eisenhardt, 2021). Matched case designs are often used in venture process studies (e.g., McDonald & Gao, 2019; Navis & Glynn, 2010; Volmar & Eisenhardt, 2025), as they allow rich accounts over long periods of time.

Our sample is three entrepreneurial ventures (see Table 1), originally identified in (*citation withheld for anonymity*) by the first author. (*Citation withheld*) sought to understand and identify how firms navigate cooperation and competition within nascent ecosystems. An interesting puzzle emerged in the course of that project: despite facing similar founding conditions, the three firms undertook radically different strategies at launch. Moreover, all three founding teams articulated ideas which – as we dug into the emerging literature on entrepreneurial theories – clearly resembled theories. As such, they offered an uncommonly well-matched sample within which to observe the interplay of theories and strategies. In contrast, the two other founding teams in the original sample did not articulate clear theories, and the ventures themselves failed fairly quickly, limiting our ability to trace the phenomenon of interest.

This matched case sample is attractive for several reasons. The three ventures were founded at the same time (2007), in the same market (California), with similar team demographics and access to funding (see Table 1). These similarities help to ensure that the founders were facing similar market forces and levels of investment. Unusually, founders at all three firms also remained in key executive roles (e.g., CEO, President) throughout the study period. This is helpful because it mitigates the possibility of major changes in theories or

strategies due to executive turnover. Finally, the three firms we study are useful in that each set of founders explicitly talked about their businesses in terms of contrarian “bets” built on overlooked or novel assumptions about the world and articulated clear organizational decisions and ecosystem configurations on the basis of this understanding. In this sense, all three articulated clear theories, which is useful because it provides an opportunity to examine how *different* theories can lead to effective strategies within a single industry context.

An important point is that all three firms achieved a measure of success, becoming among the ten largest firms in the industry and experiencing relatively consistent growth over the course of our ten-year study period. Studying successful firms limits our ability to infer causal performance implications. Yet, it also allows us to observe a variety of changes (in both strategies and theories) over time, including some that are not driven by performance shortfalls. This improves the precision and generalizability of our inducted findings.

[Insert Table 1: Sample]

## **Data collection**

We use a variety of data sources. Archival sources played a central role. Using LexisNexis and Factiva, we gathered U.S. newspaper articles about the ventures during the study period (N=886). We also manually gathered articles from cleantech internet sources such as Green Tech Media and SolarWorld (N=639), press releases issued by the ventures (N=456), published business teaching cases (N=10), books written by the founders (N=1), and blog posts written by the founders (N=39). We also captured social media posts from the founders (N=2,315). These sources provided insight into the ventures’ actions and overall strategy, as well as the founders’ perceptions and interpretation of their actions and the broader economy and industry (Table 2). Finally, we gathered archived versions of the ventures’ web pages over time,

using archive.org, which was useful in plotting the evolution of the ventures.

We supplemented these online and print sources with archival recordings of presentations and interviews by the founders (typically conducted by journalists, analysts, or academics), dating from 2007-2016. To capture these recordings, we combed video-sharing platforms such as YouTube as well as online archives and conference websites (e.g., Solar Power International). This resulted in 40 interviews and presentations totaling over 30 hours. These interviews provided insight into founders' theories and ventures' strategic actions, as well as insight into the links between the two – e.g., *why* ventures undertook specific strategic actions and *how* the founders interpreted their outcomes, often in the form of contemporaneous explanations. We also captured an additional 57 archival interviews with external informants such as partners and industry analysts, similarly dating from 2007-2016. In leveraging archival qualitative data in this way, we follow a trend in strategy and organizations research (e.g., Zuzul & Tripsas, 2020). For our study, qualitative archival data offers a way to explore how founders interpreted the landscape and put theories into action over a long study period.

[Insert Table 2: Data]

In addition to this archival data, we also conducted semi-structured interviews with firm founders and executives ourselves (N=17). These interviews were initially conducted as the primary data for the broader project on strategy and ecosystems. Here, they serve as a method of triangulating archival material and validating public statements, as well as adding detail to historical accounts. The first author conducted these interviews in two waves: first in 2014-2015 and again in 2016-2017. Interviews lasted 45 minutes to two hours and were recorded and transcribed. Interviews in the first wave covered ventures' founding and strategy up until the time of the interview, including specific actions taken (or not taken), the rationale for each, and

the informants' perception of the broader industry and ecosystem (e.g., identity and characteristics of rivals, general economic and technological trends). Interviews in the second wave addressed the same topics but probed more directly into entrepreneurial beliefs and learning processes. We also interviewed external informants (N=42), such as executives at rival and partner firms, in order to broaden our understanding of the industry and the ventures. Relative to our archival data, these interviews allowed critical insight into executives' understanding of the landscape, their rationale for actions, and how their beliefs evolved over time, often in their own words. In conducting these interviews, we employed established procedures to ensure validity, including a) interview techniques such as non-directive questioning and timeline reconstruction (Huber & Power, 1985), b) relying on informants from multiple hierarchical levels (Kumar et al., 1993), and c) ensuring informants' anonymity, which encouraged them to speak candidly.

### **Data analysis**

We began analysis by writing a chronology of events at each venture. These individual cases served to establish a timeline of events, and as a way to aggregate and synthesize our diverse and sprawling data. The cases focused on the initial founding of each venture, as well as the evolution of their strategies and subsequent strategic actions taken over time. We first identified the ventures' key activities at each point in time, as well as specific strategic actions (i.e., substantive resource commitments) taken as part of these strategies, such as product launches, market entries and exits, partnerships and acquisitions, and executive hires. We included information regarding the rationale for these actions and observed outcomes. Since we began conducting our primary interviews in 2014, a key consideration was to ensure that archival sources (e.g., online interviews) and the primary interviews converged.

We also worked to capture the founders' theories. Capturing theories in a real-world empirical setting presents a challenge for two reasons. First, while founders sometimes *explicitly* articulate theories (as do the founders at points in our study), theories are also reflected *implicitly* in founders' statements (e.g., their descriptions of the industry, or the rationales provided for undertaking a specific action). Second, there is no established method for mapping extant conceptual definitions of theories to real-world empirical settings. And, the methods typically used to capture theories in experimental work (e.g., Business Model Canvas) may not capture the founders' underlying rationale for their beliefs, or the sources of assumptions.

With this in mind, we approached each case with a broad probe, looking for insight into how founders described their business idea, the value they hoped to provide, and their understanding of how and why this idea might work. We were aware of existing definitions from the theory-based view emphasizing causal linkages as being the central and differentiating element of theories (e.g., Ehrig & Schmidt, 2022; Felin & Zenger, 2017), so we were particularly attuned to statements relating (potential) actions and outcomes. We were also aware of the literature on entrepreneurial processes, so we examined the data for related construct categories that may *inform* theories, such as analogies (Carroll & Sørensen, 2024) or identity (Zuzul & Tripsas, 2020); we also remained open to unexpected constructs (Eisenhardt, 2021).

We then moved to cross-case analysis, as is typical for this research approach (Eisenhardt & Graebner, 2007). In doing so, we began to see patterns both across the firms and over time. First, we observed the founding teams articulating both clear *problems* that they hoped to solve, as well as *solutions*, in terms of the set of organizational activities and choices they could undertake to address these problems and create and capture value. We also observed the founders articulating clear (and distinct) *logics* mapping potential solutions to problems. Through an



iterative process, we arrived at an inducted conceptualization that describes theories in terms of three interdependent elements: a) a founder's primary impetus for starting a venture (i.e., a problem to solve); b) a set of beliefs regarding *how* this problem might be addressed (i.e., a solution); and c) related beliefs about the broad economic forces underpinning *why* this particular solution will achieve the desired outcome (i.e., a logic).

The last step in our analysis was to examine patterns in terms of the interplay between theories and strategies. We had expected to be able to leverage two major sources of contrast and comparison: *within* firms (over time), and *across* firms. As we embarked on our analysis, however, we began to see patterns at the level of strategic actions as well. In particular, pooling across our data, we realized that there were important commonalities in terms of how the entrepreneurs rationalized outcomes and what they learned from the strategic actions that they undertook (see Figure 1). To capture this interplay, we first documented founders' rationale for undertaking each action in terms of specific arguments or expectations about why it was necessary and what it would accomplish. We also traced what the founders learned from each action, again using direct testimony. This allows us to capture learning in two ways: (1) as changes in beliefs that result from a given action (e.g., an updated understanding of the possibility space, often associated with new actions), and (2) through explicit statements of what was learned (e.g., "as a result we changed our thinking..."). Importantly, while founders sometimes undertook actions solely with the intent to learn (e.g. "we wanted to test..."), we also observe learning occurring without such intent.

Through this analysis, we began to better relate how theories were updated (or not) due to the learning that followed strategic actions. Here it became clear that while firm founders updated their theories based on the information generated by strategic actions, their learning was

*asymmetric*, and that not all elements of theories were subject to revision. We also began to understand how theories prompted new strategic actions. Of particular interest were the subset of strategic actions that occurred without clear setbacks, wherein executives traded smooth operations and solid performance for risky new activities and uncertain returns. Some of these changes constituted major pivots, wherein the firms undertook radical shifts in value proposition or intended customers (e.g., Camuffo et al., 2024), but in other cases they reflected additions and/or extensions to existing activities. To better understand these changes, we coded a) founders' rationale for undertaking the changes, as well as their discussion of the challenges and steps involved in executing them. We also coded b) degree of change, as a measure of the magnitude of changes needed to implement the new strategic action; this we captured in terms of new activities and capabilities required, as well as any activities or commitments that were dropped. This revealed a clear role for theories – and in particular beliefs regarding *problems* and *logics* – in surfacing and rationalizing changes both large and small.

[Insert Figure 1: Timeline of key strategic actions]

Finally, we put the preceding elements together to try to understand broad patterns in the evolution of strategies and theories over time. As insights emerged, we iterated between emergent theory and data to sharpen constructs and theoretical logic, and to ground them better in the data (Glaser & Strauss, 2017), leading to the framework presented next.

## **FRAMEWORK**

### **Founders' initial theories**

All three founding teams began with a shared understanding of the U.S. solar industry as backwards and stale, populated by undersized, high-cost local installers. Although they all identified high costs as the prevailing issue for the solar industry, the three teams interpreted the

actionable problem (i.e., the one that their ventures would aim to address) differently.

Accordingly, they formed different theories. Looking across the three ventures, however, we see that these theories, while distinct, share common elements. Specifically, each consists of a) a *problem* to be solved, b) an idea of how to organize to solve this problem and deliver value (i.e., a *solution*), and c) a *logic*, rooted in prior experience and observation of other industries, articulating why this solution would be effective (Table 3). We also observe founders attempting to validate and refine these theories prior to launch, consistent with prior work.

[Insert Table 3: Initial theories]

**Earthtech.** Earthtech was founded by three friends with backgrounds in environmental advocacy and energy; one had launched startups in the past and another had worked in the renewable energies division of a major energy firm. All three were motivated by the problem of how to grow the solar industry to address climate change. As one founder noted, “[We] hope to facilitate a massive uptake of solar electricity systems across California, and eventually the U.S.” (E1, 2008, BP). Like the other founders in the sample, they believed that incumbent installers were largely unsophisticated and unlikely to be able to scale themselves; thus, they saw facilitating growth as their unique responsibility. As one founder explained,

*“We want to see solar uptake through the middle class...I know the industry very well. I know that there [is] a deficit of strong business models that can really deliver on the mass distributed model on a large scale” (E1, 2008, NA)*

Thus, their problem to solve was how to grow the industry and deploy more residential solar, given that existing incumbents were unlikely to be able to do so. Their envisioned solution rested on the growing availability of high-quality digital satellite imagery: companies like Google and Microsoft had mapped, and made available, detailed multi-dimensional maps of many US neighborhoods, and the founders theorized that this might allow systems to be designed remotely. If so, the sales process could be unbundled: rather than an engineer/installer

needing to assess a home in person, sales and design work could be conducted remotely, from anywhere in the world. Installers, freed from having to conduct sales work themselves, could then focus on what they did best: install systems. Their envisioned solution was thus to develop this remote sales technology, perform sales themselves, and cultivate a broad network of specialist installers to do the work of installing and maintaining the actual solar hardware. As E1 explained, *“The only limitation here is digital imaging, which has millions of dollars going into it now, and we [can use] that to spread solar all over the world”* (E1, 2008, NA).

The Earthtech founders saw this solution as logical given their understanding of how industries evolved – and more broadly, how large-scale societal change was achieved. In particular, from their backgrounds in advocacy, they saw systemic change (like displacing fossil fuel generation) as arising from collaboration and coordinated action. Their understanding of their own role in coordinating installers and other industry incumbents was thus analogous to their work in rallying stakeholders around other issues. They also drew inspiration from the personal computing industry, which they understood as having taken off only because Dell was able to induce the same type of coordination: by moving sales online, Dell helped “unbundle” the supply chain, allowing manufacturers to specialize, and thus costs to fall and computers to become ubiquitous. As E1 explained at the time,

*“The whole point of Earthtech is to facilitate the scaling of solar by making it easier, not just for customers – that’s the first innovation – but also for installers and other contractors to get in the game. Our view is that to save the planet with solar we’ve got to something like 700x the installed capacity. You’re not going to do that just by growing mom-and-pop shops. With all due respect to them... that high-end, low-volume, high-cost business will always have a place for premium and elite customers, but there also has to be a Dell”* (E1, 2008, AI).

While grounded in a clear logic of the industry, Earthtech’s proposed solution was novel and built on a variety of uncertain assumptions, which the founders tried to validate before taking action. For example, to understand installers’ needs, they talked to installers and shared their idea

at sustainability conferences, where it received positive feedback. They also interviewed software developers in order to confirm that the platform could actually be built (in particular, it was unclear whether the available satellite data could produce a reliable design without more granular information about roof condition, obstructions, or other issues). As the idea continued to receive positive feedback, the founders hired a coder to prototype the technology. This confirmed that the idea was technically feasible, while also prompting them to refine their target customer from homeowners (broadly) to “small homes”, which were a better fit for the technology and more easily designed, and thus a more scalable market.

**Megasolar.** Megasolar was founded by two Silicon Valley software entrepreneurs who had sold their previous venture and were exploring new ideas. Prior to launch, the founders conducted extensive research across various industries, looking for settings where they might be able to build and then scale a high-growth business. Solar sparked their interest because it represented a chance to work in an important area, and because they saw the majority of incumbents as unsophisticated, relatively small providers that perpetuated inefficiencies and high costs. As M1 explained, *“There really isn’t a national brand, so there’s a tremendous opportunity. This is bigger than the internet”* (M1, 2007, NA). Thus, the problem they hoped to solve was how to capture market share and dominate the fragmented industry.

The founders theorized that the best way to accomplish this was to build a fully integrated, national installation business. Specifically, they believed they could use their networks to raise venture capital to acquire and merge incumbent installers; by achieving greater scale and scope in this way, Megasolar could achieve operational efficiencies and reduce its reliance on partners, leading to lower costs. Using a similar logic, the founders planned to invest in IT and operations, and to develop a lease finance product of their own. While requiring

significant up-front investment, the founders theorized that this would lead to lower costs over the long term – thus allowing them to undercut rivals and dominate the industry.

Although straightforward, this approach was fairly contrarian with respect to both the solar industry (wherein even the most successful installers operated locally) and Silicon Valley (which saw construction as an unattractive business). As M1 explained,

*“We decided to get into the hard work of actually installing solar panels on peoples' roofs. The general feedback when we were trying to raise capital was, ‘That's a bad business. It's heavy labor. There's no barriers to entry and you're not going to be able to compete against all the local players.’ That's what we were up against but we knew that was the right thing. The whole business model to make solar affordable. The only way you can make it affordable is by maintaining the cost and doing all the work yourself.” (M1, 2015, AI).*

For the founders, however, up-front investment in scale and scope followed a natural logic rooted in their experience in enterprise software. Specifically, they saw this approach as parallel to developing a software product, and then selling it across as large a customer base as possible. As M2 explained, *“The founding vision of the company was to try to bring economies of scale and a trusted brand to solar installation”* (M2, 2008, NA). M1 similarly noted,

*“Getting efficiency through [increasing] output is amazing, what it does to your cost of goods. You have your fixed part of cost of good that, if you push more volume through it, it just increases your gross margins” (M1, 2009, AI).*

To validate their idea before taking action, the founders attended conferences and talked to installers. This confirmed their view of incumbent installers as backwards and unsophisticated. As one founder explained, *“We kept asking [installers] questions about innovation. What are you doing to change? How are you accelerating adoption? And no one had any answer.”* He further recalled, *“That's my competition? Awesome. Let's get going”* (M1, 2013, NA).

**Suncorp.** Finally, Suncorp was founded by three MBA students who had met in business school, where they were on the lookout for entrepreneurial opportunities. All three had backgrounds in finance and private equity, as well as interest in climate and the environment.

They, like the other founders, saw the solar industry (and electricity generation more broadly) as economically and environmentally important, but with myopic incumbents and an outdated product that was essentially unchanged since the inception of the industry. They hoped to solve the problem of how to introduce a novel and profitable business model.

The founders' solution to this problem lay in the novel approach of *leasing* rather than *selling* solar systems to end-users. The leasing model was in use by one other firm in the industry, but had yet to find traction. The tax credits in the recently passed 2005 Energy Policy Act, however, added a new, attractive element: By raising money from investors to preemptively pay for systems, and then transferring ownership of those systems to the investors, Suncorp would be able to eliminate end-users' up-front costs, generate a 30% tax credit and reliable stream of monthly lease payments for investors, and take a cut of the proceeds for itself. And, through the tax credit, the U.S. government would be subsidizing the installation of the systems. In this sense, the Suncorp founders' solution was to capture what was essentially an untapped tax arbitrage, which would allow them to revolutionize the industry while making a healthy profit themselves. S1 explained, "*We looked and said, 'Oh, interesting, they [early entrant] really have no competition'*" (S1, 2010, AI). To accomplish this, Suncorp planned to focus on finance and partner with existing installers to actually build the systems. As one early employee and executive explained, "*[We] said, 'let's enter the parts of the value chain that have solid margins and where we think we can execute'*" (2014, PI).

The logic underpinning this solution was that innovating in business models and product designs would allow the venture to both generate new value (by expanding the market and capturing the tax arbitrage), and capture rents (by controlling access to this new and important segment of the value chain). As one founder explained, looking back,

*“When an industry disruption is in early days, analysts and investors, not to mention industry incumbents, policymakers and regulators, all seek to understand how to look at it as a business. That’s what we do best. [We] had spent years evaluating business models for value creation potential as venture capitalists, as private equity professionals, and at [business school]... We saw an opportunity to bring business model innovation to the monopoly utility market by going direct-to-consumers with solar as a service” (S1, 2016, BP).*

As this model had never been done, the founders set out to validate the idea. They first performed extensive analysis to understand the financials of the industry, and talked to colleagues with experience in the electricity industry. As their idea met positive feedback, they then began to expand their team, court investors, and develop the (novel) financial and legal infrastructure to be able to sell leases to commercial users.

**Summary.** Table 3 depicts the theories initially articulated by each founding team. Our data suggest that while all three teams formed distinct theories about the opportunities presented by the solar industry, their theories were all commonly comprised of three elements: a) the problem to solve, b) a solution for how to organize, representing a set of activities to be performed by the firm itself and potential partners, and c) a logic, often rooted in their experience and observation of other industries, for why this solution would be effective in solving their problem. We also observe all three founding teams attempting to validate these theories, such as through interviews, industry analysis, and low-cost prototypes.

### **Theories in use: Guiding action and directing learning**

Having formed and validated their theories, founders then began to take action. That is, they began to transition from a mode of testing and validation to attempting to execute the solution articulated in their theory. In this sense, theories provide a foundation for entrepreneurs to pursue a specific course of strategic action, in line with prior literature (e.g., Felin & Zenger, 2017). Yet, these actions also generate information, and in this way serve as “live tests” of the underlying theories. As expected, we observe all three founding teams updating and revising



their theories as they learn from the strategic actions taken during and after launch.

Unexpectedly, however, this learning is *asymmetric*, defined as a persistent tendency in which only a subset of beliefs is updated in light of novel information. In our setting, across all three ventures and throughout the study period, the learning that occurred pertained to solutions. Beliefs about problems and logics remained stable, even in the face of potentially disconfirming evidence, as entrepreneurs instead identified and experimented with new and alternative approaches to organizing *given* their existing problem and logic.

In many instances, theories guided strategic actions and these actions worked as expected. The founders typically interpreted these successful actions as evidence supporting their theories. For example, Megasolar's theory emphasized the importance of reducing costs by achieving scale; the founders further reasoned that the best way to achieve this scale was to acquire and integrate incumbent installers. As founder M1 stated prior to launch, "*A big part of the strategy is to do a national rollout*" (M1, 2006, NA). After some initial testing, the founders began taking actions to put this into practice. They first raised venture capital (a \$10M seed round), began identifying and evaluating installers, and (eventually) negotiated their first two acquisitions. They then worked to integrate the operations of these installers, as planned, and invested in operations software and streamlining the sales process. Broadly, they were successful in doing so. As M1 explained about 16 months later,

*"Megasolar has been able to grasp the largest market share without any technology different from its competitors, and without the brand credibility of [its three largest rivals], which have been in business longer" (M1, 2008, NA).*

Interpreting this positive outcome as further evidence in support of their theory, they continued to invest and grow, with four additional acquisitions in the next four years.

In other cases, however, founders undertook the actions prescribed by their theories but found that they did not work as planned, and were thus forced to reevaluate their theories.

Earthtech's effort to cultivate a partner network illustrates. The founders' initial theory was that they could facilitate the growth of the solar industry by using the internet to move sales and design online, thus reducing costs and allowing installers to specialize. E1 explained, "*Installers love to install, but they do not like doing sales. We are taking away the work that they do not like doing*" (E1, 2009, AI). Having tried to validate this theory (prior section), the founders committed resources to put this theory into action: they developed the remote design tool, raised investment, hired developers, and opened their website to homeowners in California. Some aspects of these actions went as expected. For example, their remote system was able to generate sound designs, and initial sales confirmed that homeowners were willing to purchase systems online (something that had never been done before). This activity also allowed Earthtech to refine its estimates of cost savings, demonstrating a savings of around \$2-3k/sale, or 10% of the total cost. Yet, the founders also ran into challenges that their pre-launch testing had not revealed. Specifically, while installers had been willing to install a few "one off" Earthtech systems, they were largely unwilling to adopt the tool at scale, despite the savings it allowed. Because cultivating an installer partner network was central to Earthtech's theory, its rejection by installers was unexpected, and a major setback.

Yet, setbacks of this nature were not uncommon: Across our pooled data, we observed multiple instances of the founders taking actions in line with their theory but then seeing expected outcomes fail to materialize. These failures, like straightforward successes, generated information from which the founders could (and often did) learn. Yet, how the founders interpreted these setbacks was patterned and asymmetric, in that they learned about their solution (how to organize), while reaffirming their beliefs regarding the problem and logic (Table 4).

[Insert Table 4: Examples of asymmetric learning from strategic actions]

Earthtech's founders recognized that failing to achieve buy-in from installers was a major problem. Yet, they did not question whether buy-in *could* (or should) be achieved. Instead, they rationalized the unexpected failure with an assessment that Earthtech's technology was underdeveloped, and that while installers were unwilling to use the platform *as it was*, improving the platform would increase its appeal. This prompted the decision for Earthtech to enter installation to prove the platform worked. E2 later explained, "*The remote solar design piece was a technical challenge, and none of us were technical people*" (E2, 2014, PI). E1 also noted,

*"We [realized we] had to ground proof and test out the remote solar design to make sure it would work. So we started with this lab and ran crews for about a year, which allowed us to hone the tool and make sure that the installation systems were feeding back into the design process"* (E1, 2014, PI).

In this way, the Earthtech executives explained the unexpected failure (i.e., their initial attempt at cooperation had failed because the technology was inadequate) *and* identified a viable path forward (i.e., internalize installation to improve the tool), without revising other elements of the theory (i.e., co-specialization is a sound approach for growing the industry).

Suncorp's entry into sales provides another example. The founders' initial theory had envisioned building a profitable firm by introducing a disruptive new technology (lease financing) and organizing as a B2B provider of lease financing. By providing a valuable product (i.e., tax equity leases) in a market with minimal competition, the founders believed that Suncorp would be able to control the most profitable activities, while outsourcing the less scalable and less profitable sales and installation activities. Guided by this theory, Suncorp's founders developed a lease, partnered with ten incumbent installers (who, as above, would perform sales *and* installations internally), and secured major investments from large banks and other commercial entities to provide the capital to finance the installations. This set of strategic actions was initially effective, and as of 2009 Suncorp was one of the fastest growing firms in the

industry. By 2010, however, it became clear that the industry was shifting, and that the founders had substantially overestimated the defensibility of finance: rival firms were entering the market, creating price pressure which pushed margins from 20% in 2008 to under 9% in 2010.

In light of this new information, Suncorp's founders retained their initial belief regarding the problem to be solved (to innovate and profit in a stale market) as well their logic for why their entry into finance had worked (profits accrue to firms that control attractive markets). Their learning instead was directed toward their understanding of what this meant in terms of the corresponding solution. Specifically, they came to see *sales* as the important bottleneck, and thus the main driver of market power and profitability in the future. This meant that given their *existing* understanding of problem and logic, Suncorp needed to perform sales itself. This in turn required building out their own internal sales operation, hiring staff, building a consumer brand, and developing system design software. As S1 explained at the time,

*"Ultimately, [finance] will become more and more commoditized, and that's why the other things we're building – software systems to bring down transaction costs, smart pricing models, building a consumer brand, getting the best distribution – are going to be the better long-term value" (S1, 2010, NA)*

Entering sales represented a substantive change from Suncorp's initial set of activities as a B2B finance provider, but was consistent with the founders' problem, logic, and updated understanding of where the best opportunities for innovation and profit were. As S2 explained,

*"Three-fifths of the cost of residential solar is soft costs. Customer acquisition, municipal permitting, and things like that. The real winners in the marketplace will be those who can minimize those costs" (S2, 2012, NA)*

**Summary.** Overall, a key insight is that actions taken to implement a theory generate information, which might then be used to assess the validity of that theory. Yet, what executives and founders learn from these tests is *asymmetric*: they consistently interpret unexpected outcomes as revealing new information about their proposed solution and how best to organize,

*without* revising beliefs regarding the problem to be solved or the underlying logic linking problems and solutions. A key insight is that (as in science), the failure of a hypothesis may stem from a variety of factors, and what the entrepreneur as scientist learns is a matter of judgment and interpretation. What entrepreneurs need to do, however, is to provide a proximate explanation for negative outcomes, while also identifying a path forward from among the set of possible options (Packard et al., 2017). Learning about *how to organize* provides such an explanation, reducing dissonance as executives interpret unexpected information as an input to further theorizing (Burgelman & Grove, 1996; Ott & Hannah, 2024).

Beliefs about *how to organize* are more likely to be revised than beliefs about the *problem to solve* or *logic* for two reasons. First, beliefs about how to organize (solutions) reflect an understanding about the appropriate set of actions and activities *given* a particular problem to solve and a logic dictating which solutions map to that problem. In this way, beliefs about problem and logic precede solutions. Moreover, solutions necessarily comprise a *set* of activities and associated beliefs (Ehrig & Schmidt, 2020; Felin et al., 2024). For example, the Megasolar's founders' beliefs regarding scope and scale informed their entry into installation, sales, and finance, while also guiding a competitive posture to incumbent installers. Theory on managerial cognition describes *core beliefs* as beliefs that drive multiple peripheral beliefs, such that they are more deeply embedded and central to an actors' mental representation (Eden et al., 1992; Lyles & Schwenk, 1992) – here, a theory. Because dense webs of connections inhibit change, the interconnectedness of “core” elements may make them more inert than the peripheral beliefs that build upon them (Hannan et al., 1996; Kiss & Barr, 2015). Rather than being discarded in the face of negative information, core beliefs are instead likely to shape where the search for alternatives takes place (Cyert & March, 1963; Nadkarni & Narayanan, 2007) – i.e., in

alternative solution possibilities that fit existing problem and logic beliefs.

Second, relative to beliefs about organizational activities, beliefs about the problem to solve and underlying logics are more likely to be rooted in (or even distilled from) prior experience (Felin & Zenger, 2017; Gavetti & Menon, 2016) and/or founders' values and identity (Rindova & Martins, 2018; Zuzul & Tripsas, 2020). Indeed, a striking pattern across all three founding teams was the degree to which they described their own (idiosyncratic) problems and logics as “common sense” given their background – e.g., Earthtech's activist founders attempting to use coordination and cooperation to resolve a pressing global environmental issue, or Suncorp's elite MBA founders linking profitability to the selection and execution of attractive markets. Beliefs linked to values and identity are forged and reinforced through experience over time, and are understood to be deeply held, with powerful impacts on cognition and action (Higgins, 2016; Rindova & Martins, 2018; Schwartz & Bilsky, 1987). When the external environment impinges on deeply held beliefs, individuals react in creative ways to maintain these beliefs (Elsbach, 2009; Pratt et al., 2006). Revising solution beliefs provides one such means, leaving problem and logic beliefs untouched. Similar arguments have been advanced by philosophers of science (Lakatos, 1970; Popper, 1959), who argue that when new information (e.g., experimental results) threatens the central beliefs on which a theory rests, actors may invent alternative hypotheses, which then “*bear the brunt of tests and get adjusted and re-adjusted, or even completely replaced, to defend the thus-hardened core*” (Lakatos, 2014:191).

In line with this argument, we do see founders continuing to engage in formal experimentation after launch. Yet, these experiments did not test core beliefs regarding problem and logics. Instead, they focused on validating peripheral beliefs around how to implement specific solutions. For example, having decided to enter sales (above), the Suncorp founders set

out to test the efficacy of different ways to do so. As one executive explained,

*“We started with a small group to pilot [inside sales] and to build that skill set of how we sold over the phone. As you might imagine, we got ten people and they started selling over the phone, and we did some things wrong and did some things right and kind of learned...about who's good at that, and the technology they need and all that kind of stuff.” (2014, PI)*

Similarly, Earthtech’s founders – maintaining their belief in the value of cooperation – continued experimenting with ways to coordinate with specialist installers. E2 explained,

*“There’s been experimentation around the edges. Like delivering the permit to the permitting office. Is that a truck run that we want them to do because they know the guy and they have beers with him, or is it better to just have our certified overnight system that we can control?” (E2, 2014, PI).*

Across all three firms, we see multiple instances of post-launch experimentation, all focused on how to best execute specific elements of their general solution concept.

### **Theories in use: Surfacing and justifying novel opportunities**

Research on entrepreneurial strategy formation depicts strategy as emergent and adaptive, as ventures’ actions reveal unexpected challenges and resolve uncertainties (Kirtley & O’Mahony, 2023; McDonald & Gao, 2019). Consistent with this view, we see the firms adapting their strategies (and their solution concepts) in response to unexpected setbacks. Yet, we observe changes occurring without clear negative feedback as well. This is surprising, given our preceding finding that core elements of founders’ theories are largely resistant to revision, as prior research argues that inertial mental models can blind entrepreneurs to novel opportunities or changes in the environment (Barr et al., 1992; Tripsas & Gavetti, 2000). Moreover, firms accumulate resources, routines, and stakeholder relationships over time, which might be expected to further cause firms’ strategies to stabilize over time, absent performance shortfalls (Gavetti & Rivkin, 2007; Posen et al., 2018).

Digging deeper, we see that rather than driving strategic inertia, the stable elements of

founders' theories may actually set the stage for change by sensitizing founders to novel opportunities and helping them consider novel solution concepts that align with those beliefs. In this sense, the stable elements of founders' theories may enable them to generate novel actions by providing a foundation for imagining possible options to achieve the desired outcome (Packard et al., 2017). We observe two mechanisms, with founders' (stable) beliefs a) *surfacing* new ways to organize, and b) *justifying* the potentially costly pursuit of these opportunities, by allowing founders to envision and articulate the value of the prospective change. We also observe that, as a function of the teams articulating different theories, founders were sensitized to (and willing to pursue) different opportunities, leading to strategic divergence over time.

[Insert Table 5: Stable beliefs facilitating strategic change]

Earthtech's 2014 launch of an in-house startup "incubator" illustrates. The founders' initial theory emphasized growing the solar industry by improving coordination and facilitating specialization among incumbents. This called for internet-enabled design and sales, and cultivating a network of partners in installation and finance. Over the next few years, this theory and strategy paid off: By 2013, the firm had \$120M in annual revenue, growth rates of 60% per year, over 110 partnerships in installation and sales, and was viewed as a leader.

Yet, in line with their emphasis on growing the industry by coordinating collective effort, the founders were also on the lookout for opportunities to help a broader set of actors. As a result, they began to consider the high barriers that potential entrants to the industry faced – the barriers that they had overcome. E1 explained, "*having done it once myself, I know how [expletive] hard it is*" (E1, 2014, PI). And, in line with their existing problem to solve and logic, they began to theorize about alternative actions that Earthtech might take. E1 explained,

*"You need, like, thousands of companies having a crack to fill out residential solar, commercial solar, ground mount solar, all the smart pieces that go with*



*it, all the software, and storage, and stuff that makes all the solar actually become 10% of generating capacity. For that you need entrepreneurs having a go. [We need] to support those solar entrepreneurs to succeed” (E1, 2014, AI).*

This theorizing eventually surfaced a novel solution – i.e., a new set of activities that would better allow Earthtech to solve its focal problem of scaling the industry. Specifically, the founders envisioned launching an in-house incubator that could support fledgling solar entrants. E1 explained, it could be *“an attempt to spawn entrepreneurial efforts and support them to succeed at a higher rate than they would in the wild”* (E1, 2014, PI).

Executing this addition to strategy would be resource intensive, however. It would require executive leadership, and they would also need to allocate office space, build a team, and set aside resources to provide portfolio ventures with capital, mentoring, and legal and public relations support. It was also a highly unusual activity for a firm like Earthtech to undertake: the first cohort of ventures, admitted in 2015, focused on a range of activities, including remote sales, energy monitoring, and crowd financing for renewable energy projects. Many of these firms competed directly with Earthtech, or provided services for its competitors. E1 explained, *“These guys are sourcing appointments for our competitors. I think that’s [expletive] great because they could make a business out of that”* (E1, 2014, PI).

However, the founders were able to justify these costs, both internally and to external stakeholders, given their understanding of how to grow the industry. E1 explained,

*“If we succeed in our task, we’ll have thousands of solar industry workers here — they’ll want to walk to work or cycle. They’ll become the population that helps make [the global transition to solar] happen. The whole thing will be this nice, synergistic, lift-all-boats kind of deal” (E1, 2014, NA).*

Megasolar’s entry into solar panel manufacturing offers another example. The Megasolar founders’ theory emphasized dominating the fragmented industry (the problem to solve) by bringing economies of scale and scope to reduce overall costs (the logic). At launch, this implied

entering sales, installation, and finance, which they achieved by acquiring and integrating incumbent installers (the solution). By 2013, Megasolar had become one of the largest installers in the country, and had been able to achieve substantial cost reductions, with internal documents showing costs having fallen more than 10% per year from 2011 to 2013.

Megasolar had achieved these reductions through operational improvements and tight integration of its sales, installation, and financing operations. Like almost every other firm in the industry, it had not attempted to bridge into actually making photovoltaic panels. This stemmed from the common understanding that *installing* and *manufacturing* panels required vastly different capabilities, and thus presented little opportunity for effective synergy. Moreover, panel manufacturing was viewed as an unattractive business: capital intensive, low-margin, heavily dependent on government subsidy, and haunted by high-profile failures like the 2011 Solyndra collapse.<sup>1</sup> Panels were also readily available and becoming more so: as Chinese panel production scaled, prices had been falling – more than 83% over the preceding seven years.

The prospect of entering panel manufacturing was thus considered truly contrarian. Yet, consistent with their focus on economies of scale and scope, Megasolar's founders were especially prepared to surface opportunities to expand into new areas that could yield benefits in terms of operational efficiency. In 2012, this led them to theorize that there was a new opportunity to reconfigure Megasolar's installations to be more efficient if it designed its own panels, since panels and racking could then be more tightly integrated and its installation operations could be optimized to work with these specific products. M1 explained, "*The only way to get to scale is if you control and maintain your own destiny*" (M1, 2014, NA).

In this way, the founders' emphasis on scope and scale economies primed them to surface

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<sup>1</sup> In fact, by 2013, only one firm manufactured *and* installed panel hardware, and this was a large manufacturer that had diversified from hardware to build a dedicated installation operation for its own panels.

a novel opportunity. It also allowed them to justify the move to external stakeholders despite heavy skepticism. As M1 explained to a financial analyst in a 2014 interview,

*"If you look at the history of the company, we've continued to convert it to integrated. We started off with sales and marketing, and we've vertically integrated our installations. Then, we vertically integrated our financing. Then, we vertically integrated our mounting hardware. By doing it that way, we've been able to provide a cost structure that's the most competitive in the market space which allowed us to increase our market share. The Megasolar way is vertical integration to reduce the cost of energy." (M1, 2014, AI)*

The massive cost and disruption to very efficient existing activities was framed as logical, given what their theory meant for the future. As M2 explained,

*"Given that there is excess supplier capacity today, this may seem counterintuitive to some who follow the solar industry. What we are trying to address is not the lay of the land today, where there are indeed too many suppliers, most of whom are producing relatively low...efficiency solar cells at uncompelling costs, but how we see the future developing" (M2, 2014, BP).*

***Extension to the analysis: Rejecting opportunities.*** Theories do more than facilitate the pursuit of novel opportunities – they may also prompt entrepreneurs to reject presented opportunities when seen as inconsistent with problem and logic beliefs.<sup>2</sup> Capturing changes that do not occur is notoriously difficult, as there is typically no trace to observe (Kirtley & O'Mahony, 2023). While we cannot capture the full set of rejected opportunities, a comparative case design helps by allowing us to observe similar decision points that the three firms faced, but which led to different actions taken. For example, in 2009 all three firms were offered an opportunity to participate in a cooperative, industry-wide marketing effort similar to the "Got Milk?" campaign. Megasolar's founders – whose theory emphasized firm scale to drive down costs and dominate rivals – opted not to participate. M1 explained at the time, *"I think it will still be individual efforts establishing a brand within [each company]... I like the idea – I think it will*

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<sup>2</sup> We thank a reviewer for making this observation.

*be a long time until we get there*” (2009, NA). In contrast, Earthtech executives presented with the same choice, jumped at the opportunity to advance the collective good. One founder later rationalized, *“We’ve always had a lift all boats kind of mentality. Put back into the general effort to increase the market. That will behoove us as well”* (E1, 2014, PI).

Founders’ theories similarly prompted different decisions with respect to new state markets. Again emphasizing scale to drive down costs and beat rivals, Megasolar entered new markets early. One executive explained, *“We took a big bet on [Southern state] before the rebate program existed, and literally spent a lot of money building our warehouse, building marketing, building partnerships”* (2014, PI). In contrast, Suncorp’s founders, who hoped to build a profitable firm by picking attractive markets, passed on the same opportunity. One founder explained, *“You have to make sure the economics are there where you can price [given] prevailing energy rates”* (S1, 2015, AI). As these examples illustrate, just as founders’ theories help to surface and justify substantive strategic changes, they may also guide which presented solutions get *rejected* – and help justify these decisions to stakeholder groups.

**Summary.** Overall, our data suggest that founders’ stable beliefs regarding problem and logic facilitate strategic change in two specific ways. First, they help entrepreneurs surface novel opportunities, even in the absence of negative feedback. That is, rather than preventing executives from perceiving novel solutions and thus constraining a path of continued investment and commitment to existing activities (Cyert & March, 1963; Tripsas & Gavetti, 2000), stable elements of theories may serve as catalysts for a more generative mindset – sensitizing executives to particular changes outside of their firm and thus allowing serious consideration of novel solutions (Eisenhardt & Bingham, 2017). Indeed, recent work suggests that some cognitively sophisticated strategists combine existing knowledge and imagination to generate

possibilities beyond what others have envisioned (Packard et al., 2017; Rindova & Martins, 2021; Rindova & Martins, 2023; Rindova & Martins, 2024). In this way, founders work to create a more favorable environment for their activities and look for ways to improve their odds of success (Gambardella & Messinese, 2025). Our findings highlight the fact that this effort to craft novel possibilities continues beyond founding, as founders' problems to solve and logics prompt them to engage in questioning that surfaces innovative alternatives for how to organize.

Second, stable beliefs regarding the problem to solve and logic can also help founders justify *acting* on novel discoveries, even when doing so imposes costs in the near term. Specifically, by allowing founders to envision and compare alternative futures (e.g., Megasolar with and without access to panels), and by providing an internal logic to understand the implications of these alternatives, the stable elements of theories may make an opportunity more "real" and thus worth pursuing. In the same way, theories may also allow earlier decisions to be (re-)contextualized, allowing actors to interpret them as incremental steps justifiable in the pursuit of a grander goal. As such, theories may decrease psychological attachment to existing activities, thus reducing the perceived cost of change (Sleesman et al., 2018; Staw, 1981). Indeed, Earthtech explicitly framed its own actions as incremental, with the founder explaining, "*Our ultimate dream is to make a significant contribution towards climate change, but for now we're taking these initial baby steps*" (E1, 2008, BP). In this way, theories make it easier to *justify* pursuing changes (or not), increasing founders' willingness to overcome existing routines and helping manage stakeholders through change (McDonald & Gao, 2019).

Intriguingly, because each founding team was guided by a unique theory, we also see the teams continually surfacing, justifying, and rejecting *different* opportunities, despite being exposed to similar environmental changes. For example, while Earthtech's founders were finding

new ways to build out a vibrant partner ecosystem and support collective growth, Megasolar's founders were consistently pursuing ways to expand their vertical and horizontal scope, in order to reduce their own costs and capture market share. The stability of *problem* and *logic* beliefs thus did more than just drive heterogeneity across the firms (Felin et al., 2024) – it also introduced a degree of patterning in the options that each pursued (Packard et al., 2017). Indeed, executives from all three firms identified this thread – framing (even potentially dramatic) strategic changes as united by a common understanding of how (and why) they had chosen their strategy. For example, Suncorp's founder explained,

*"A lot of people will look at [this entry into installation] and say, "This was a strategy shift." But it really wasn't a strategy shift. The strategy from the very beginning of the company was, let's figure out how... to own the pieces of the value chain that have scale advantages." (S1, 2015, AI)*

In this way, the stable elements of founders' (idiosyncratic) theories acted to sustain and even enhance heterogeneity between the firms' strategies as each adapted over time.

## DISCUSSION

In this study, we examined how three different firms in the same nascent industry translated distinct entrepreneurial theories into strategy over time. Our empirical analysis allows us to induct a model of theory *enactment* – i.e., how theories guide action, and how these theories are then updated (or not) on the basis of the new information generated by these actions. This model begins by disaggregating entrepreneurial theories into three related subsets of beliefs: a) the *problem* to solve, b) how to organize to deliver a *solution* to that problem, and c) the underlying *logic* linking problems and solutions. In line with existing research (e.g., Agarwal et al., 2025; Camuffo et al., 2024), we observe founders testing their theories prior to launch through activities such as interviews, market analysis, and prototyping. Yet, they also *continue* to learn after launch, as they a) take actions and make strategic commitments guided by their

theories, and b) learn and update their theories based on the novel information generated by these actions. In this sense, their strategic actions constitute “live tests” that shed light on the viability of their theories (e.g., Pillai et al., 2020; Zellweger & Zenger, 2023).

When founders take actions on the basis of their theories, they expect specific outcomes to occur. When these outcomes do occur, founders interpret this as evidence validating their theory. In contrast, when expected outcomes do *not* materialize, founders must interpret *why* this occurred, and posit a (new) path forward. A key insight and central aspect of our inducted framework is that the learning that results from unexpected occurrences is asymmetric: founders consistently revise their beliefs about how to organize (i.e., solutions), while holding beliefs about the problem and logic unchanged. We term this *asymmetric learning*. Our analysis further suggests two factors that contribute to asymmetric learning, neither of which are reflected in prior research: a) the degree to which beliefs are core vs. peripheral within the theory, and b) the source of the beliefs and their link to founders’ values, identity, and experience.

Finally, we unpack a surprising juxtaposition in that stable beliefs, rather than driving strategic stasis and decline, may actually allow for continued adaptation over time – and continued heterogeneity in strategies. This effect arises because stable beliefs regarding problem and logic may help a) *surface* novel solutions for (existing) problems given founders’ (existing) logics. They may also b) help executives *justify* pursuing these novel opportunities to both themselves and to stakeholders, even when doing so might impose substantial costs in the short term. Thus, whereas prior work often associates stable beliefs with strategic inertia (e.g., Barr et al., 1992; Tripsas & Gavetti, 2000), we see that theories can be *generative*, and may facilitate continued exploration rather than local optimization. These same stable beliefs may also prompt founders to *reject* novel solutions that are seen as inconsistent. Collectively, these effects may

reinforce initial differences between firms, and preserve strategic heterogeneity even in the face of common isomorphic forces (See Figure 2).

[Figure 2: Propositions for a model of theory enactment]

Overall, this model extends research and theory on the theory-based-view and entrepreneurial strategy formation by providing what we believe is one of the first studies of the unfolding interaction between theories and strategic actions over time.

### **Towards a richer operationalization of entrepreneurial theories**

Entrepreneurial theories have been conceptualized as a mental model of a business idea; they articulate a potential future, often as a problem and a corresponding solution, and the various assumptions, contingencies, or actions necessary for this idea to be realized (Camuffo et al., 2024; Ehrig & Schmidt, 2022; Felin & Zenger, 2017; Wuebker et al., 2023). Empirical studies (e.g., Camuffo et al., 2024; Novelli & Spina, 2024) in turn typically rely on templates like the Business Model Canvas (Osterwalder & Pigneur, 2010), which capture the value proposition and key elements of the corresponding solution (e.g., activities, revenue streams, customer segments). Although this approach enables quantification of changes in specific hypotheses, as well as the subsequent impact on performance, this approach offers less insight into the underlying logic and relationships between the beliefs that comprise a theory – i.e., the critical causal understanding of *why* solutions solve problems. Our inductive conceptualization operationalizes theories in terms of three interdependent elements (i.e., problem, solution, and logic). In doing so, we contribute to a reinvigorated emphasis on the role of these causal underpinnings, which have been much theorized (Felin & Zenger, 2017) but are rarely afforded a starring role, especially in empirical work. We also add nuance to the relationships between problems and solutions *within* entrepreneurial theories.



One important implication of this operationalization is to highlight the internal structure of theories, and in particular the distinction between core vs. peripheral beliefs. While early literature on the theory-based view rarely differentiated among the beliefs that comprise theories (e.g., Felin & Zenger, 2009), recent conceptual work implies that beliefs exhibit relatively complex relationships, potentially differing in their logical ordering (i.e., subproblems nested within problems) or degree to which they are “core” to the theory (Ehrig & Schmidt, 2022; Felin et al., 2021). Recent empirical work also (implicitly) depicts some beliefs as more or less foundational, with some belief changes constituting “radical” pivots (Agarwal et al., 2025; Camuffo et al., 2024). Yet, what this distinction means in practice, or *why* some beliefs are core or not, is often fuzzy. We show that core beliefs around the problem and logic provide internal consistency across otherwise disparate beliefs, allowing actors to weave together *unique* but subjectively *rational* (Packard et al., 2017; Ryall, 2003) worldviews, often built on their idiosyncratic prior experience. Importantly, this is different than founders assigning different priors to possible outcomes: two entrepreneurs with similar theories (as captured by a Business Model Canvas) yet different underlying core beliefs might respond to the same stimuli differently and develop unique patterns of action as they search for a viable strategy, rather than a universal “truth” (Zellweger & Zenger, 2023). The presence of core beliefs prompts interesting questions for future work. Core beliefs (e.g., the role of economies of specialization vs. scale in driving costs) may be less directly testable than the peripheral beliefs they support. To what extent can entrepreneurs test these core beliefs – and if so, how?

Our tripartite operationalization also highlights the possibility of a richer set of relationships between problems and *multiple* associated solutions. Extant theorizing focuses on how entrepreneurs can better understand the viability of a *given* problem-solution pair (i.e., using

experiments to identify unfounded assumptions or logic to reveal inconsistencies) (Ehrig & Schmidt, 2022; Zellweger & Zenger, 2023). What is missing is the generation of the broader set of alternatives from which this pair is selected. Indeed, recent conceptual work argues that theories may be generative, spurring novel (re)combinations and the identification of unrealized futures (Rindova & Martins, 2024). But what might this look like in practice? Our operationalization brings the generative potential of theories to the forefront, highlighting the possibility of a “one to many” relationship between a problem and *various* potential solutions for which a logic holds. Selecting solutions from this set – and even subdividing the problem into subproblems – may not be a purely logical exercise as depicted in prior work (Felin et al., 2021), but rather one in which the entrepreneur exerts substantial creativity and agency. This also implies that elements of theories (and in particular problems and logics) may be preserved across pivots. Indeed, an intriguing pattern in our data is that changes described as pivots by *external* observers were not perceived as such by the founders themselves. Research on theory-driven pivots (e.g., Valentine et al., 2024) may benefit from exploring this discrepancy, as well as how theories might allow entrepreneurs to signal adaptability (or alternatively, consistency) to stakeholders (McDonald & Gao, 2019; Zuzul & Tripsas, 2020).<sup>3</sup>

### **Towards a longitudinal understanding of entrepreneurial theories and strategy**

We additionally unpack how theories are translated into strategies *over time*, with implications for strategic heterogeneity and competitive advantage. Work on the theory-based view argues that entrepreneurial theories are a source of novel or “great” strategies and competitive advantage, parallel to alternative perspectives such as those that focus on resources,

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<sup>3</sup> This observation likely offers important methodological insights for the study of pivots as well, as researchers may benefit from capturing pivots through both entrepreneurial self-assessments (e.g., Valentine et al., 2024) and external stakeholder assessments (e.g., McDonald & Gao, 2019). More broadly, how entrepreneurs are asked about pivots (vs. changes) is likely to affect the nature of the insights gleaned.

position, or practice (e.g., Jarzabkowski, 2004; Porter, 1996; Wernerfelt, 1984). Yet work on entrepreneurial strategy more broadly would suggest that the realization of this advantage hinges not just on *forming* an effective theory, but on *translating* this theory into a set of activities and organizational choices (Ott & Eisenhardt, 2020; Wuebker et al., 2023), as well as flexibly *adapting* the theory and strategy to new opportunities and setbacks as they arise (Bingham, 2009; Rindova & Kotha, 2001). Our contribution is in tracing the dynamics of theories and their translation into strategy, to provide insight into how theories can help balance the task of adapting as industries evolve (Barr et al., 1992; Tripsas, 2009) with maintaining internal fit around what made a strategy effective in the first place. In doing so, we contribute to research on the theory-based view and connect to literature on mental models and strategic inertia.

In the “entrepreneur as scientist” analogy, theories are depicted as readily updatable in the face of novel information (e.g., Camuffo et al., 2024; Zellweger & Zenger, 2023). Yet, the focus of this research is on the earliest stages of the entrepreneurial process, when cognitive and organizational plasticity is highest (Gavetti & Rivkin, 2007; Nelson & Winter, 1982). We shift our lens to examine the later process of how entrepreneurs take action and learn from these actions over time. In doing so, we see that learning within theories from strategic actions is unexpectedly asymmetric, and that the beliefs that comprise theories are not uniformly subject to scrutiny and revision. This observation resonates with recent work that emphasizes underappreciated challenges in interpreting and integrating feedback (e.g., Ott & Hannah, 2024; Zellweger & Zenger, 2023) – an essential step that has been largely overlooked in extant theorizing. One implication is the need for entrepreneurs to be thoughtful in how they *update* their theories, in addition to how they form and test hypotheses, particularly given the normative intent of the theory-based literature (e.g., Felin et al., 2021; Felin et al., 2024).

Our depiction of theories as not readily (or uniformly) updatable bridges the literature on the theory-based view with a rich history of work in strategy that unpacks the relationship between cognitive and strategic change (or inertia). Historically, stable beliefs (as we observe in terms of problems and logics) have been equated with strategic stasis and declines in performance (e.g., Barr et al., 1992; Tripsas & Gavetti, 2000). Recent theorizing offers a more nuanced view, arguing that stable beliefs (often in the form of a founders' identity) may be associated with either a predisposition for inertia or change (e.g., Grimes, 2018; Zuzul & Tripsas, 2020). Zuzul and Tripsas (2020), for example, show that founders who see themselves as “discoverers” are consistently more willing to adapt their products and business models, relative to those who identify as “revolutionaries”. In contrast, we shift from which *founders* are likely to change (Fauchart & Gruber, 2011), to how their *ideas* change. In doing so, we see that entrepreneurs retain substantial freedom to adapt their strategy, even when portions of the theory remain stable – which prior work might characterize as cognitive inertia.

This seeming incongruity highlights the hidden duality of stable beliefs. Prior work argues that theories are beneficial when, and in fact, *because*, they are built on contrarian beliefs, which lead to novel strategies (Felin & Zenger, 2009; Wuebker et al., 2023). How is this uniqueness sustained, in the face of isomorphic pressures and shared competitive environments (DiMaggio & Powell, 1983) where once contrarian ideas might be forsaken, or become the status quo? Our analysis suggests that the answer lies in the unique stable beliefs of the founders, which, due precisely to their stability, serve as a pillar on which idiosyncratic strategies are renewed. In this sense, stable beliefs not only reinforce past behavior but also prompt entrepreneurs to undertake novel strategic actions toward the pursuit of an envisioned future state (Kelly & Amburgey, 1991; Rindova & Martin, 2024). Thus, the link between cognitive and

strategic inertia may be more nuanced than often credited: While cognitive inertia *may* prompt strategic inertia (i.e., stasis on existing activities), stable beliefs may also drive strategic *momentum* (i.e., patterned and predictable change) (Jansen, 2004). In this way, theories may help firms sustain competitive advantage by supporting *continued* strategic change in ways that competitors are unlikely to see and/or unwilling to undertake. This may even involve dispensing with firms' own existing activities in favor of superior alternatives, thus allowing for a better allocation of scarce resources (Jannace & Camuffo, 2023).

As in all research, aspects of our study offer opportunities for future research, as well as important boundary conditions. One key design choice is that we study three relatively successful firms. Doing so allows us to trace the interplay of theories and strategies over a long period (ten years), but limits our ability to theorize on the performance implications of the dynamics we observe. Future research might examine whether strategic momentum might drive firms along a *failing* course. A related boundary condition is that we study a single industry that exhibited significant growth as well as dynamism throughout the study period. These features allowed a myriad of novel (and varied) business models and ways to succeed, as well as underlying theories. Our framework may not apply in less forgiving environments, or in the case of massive shocks that overwhelm firms' ability to adapt (e.g., Zuzul & Tripsas, 2020). A particularly compelling question for future research is what happens when founders' underlying logic is "out of step" with the actual economic forces of the industry. Finally, an important consideration is that we sample three firms whose founding teams held clear theories. While this makes sense given our interest in the interplay of theories and strategy over time, not all entrepreneurs necessarily hold (or act on) well-articulated theories (Felin & Zenger, 2017). When theories are more weakly held, do firms become more inertial, due to an inability to surface

novel opportunities (e.g., Tripsas & Gavetti, 2000)? Or more dynamic, with greater opportunism and (potentially) less cohesion (e.g., Agarwal et al., 2025; Tucci et al., 2025)?

## **CONCLUSION**

While a rich body of work underscores the importance of forming and testing entrepreneurial theories, we know less about how the interplay of theories and strategy unfolds over time. Through an in-depth longitudinal case study of three ventures over ten years, we develop a model of how theories are enacted – i.e., translated into action and then updated (or not) over time. Our model contributes by illuminating a surprising juxtaposition, with entrepreneurs updating only selective elements of their theories, and by uncovering how theories – even when resistant to change – may nonetheless enable entrepreneurs to surface and justify novel opportunities over time. Overall, we hope to provide a richer understanding of the continued role of entrepreneurial theories over the life of the firm.

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## TABLES AND FIGURES

**Table 1: Sample ventures and founders**

| Venture   | <u>Venture Details</u> |                |                 |                 | <u>Founder Details</u> |               |              |                |                |                |
|-----------|------------------------|----------------|-----------------|-----------------|------------------------|---------------|--------------|----------------|----------------|----------------|
|           | Year                   | Initial Market | Initial Funding | Funding Sources | Founder Code           | Age at Launch | Startup Exp. | Highest Degree | Prior Function | Prior Industry |
| Earthtech | 2007                   | West Coast     | \$2.5M          | VC, Angels      | E1                     | 36            | No           | BS             | Gen. Mgmt      | Advocacy       |
|           |                        |                |                 |                 | E2                     | 39            | Yes          | MBA            | Gen. Mgmt      | Advocacy       |
|           |                        |                |                 |                 | E3                     | 36            | No           | MS             | Finance        | Energy         |
| Megasolar | 2007                   | West Coast     | \$10M           | VC, Self        | M1                     | 30            | Yes          | HS             | Gen. Mgmt      | Software       |
|           |                        |                |                 |                 | M2                     | 32            | Yes          | BS             | IT/Engin.      | Software       |
| Suncorp   | 2007                   | West Coast     | \$12M           | VC, Angels      | S1                     | 28            | No           | MBA            | Gen. Mgmt      | Private Eq.    |
|           |                        |                |                 |                 | S2                     | 30            | Yes          | MBA            | Analyst        | Private Eq.    |
|           |                        |                |                 |                 | S3                     | 30            | Yes          | MBA            | Finance        | Finance        |

**Table 2: Count of interviews and archival materials**

|                                     | <u>By year</u> |     |     |     |     |     |     |     |     |      | <u>By venture</u> |       |      |
|-------------------------------------|----------------|-----|-----|-----|-----|-----|-----|-----|-----|------|-------------------|-------|------|
|                                     | '07            | '08 | '09 | '10 | '11 | '12 | '13 | '14 | '15 | '16+ | Earth.            | Mega. | Sun. |
| Primary interview                   |                |     |     |     |     |     |     | 9   |     | 8    | 3                 | 5     | 9    |
| Archival interview                  | 1              | 2   | 3   | 2   | 4   | 1   | 3   | 4   | 9   | 3    | 7                 | 15    | 10   |
| Archival presentation               |                | 1   |     |     | 2   | 1   | 3   | 1   |     |      | 8                 | -     | -    |
| Blog post                           |                | 8   | 2   | 1   | 5   | 1   | 6   | 5   | 3   | 8    | 23                | 10    | 6    |
| Firm press release                  | 11             | 7   | 3   | 37  | 59  | 6   | 74  | 68  | 119 | 72   | 44                | 287   | 122  |
| News article                        | 8              | 67  | 77  | 79  | 126 | 87  | 142 | 212 | 345 | 298  | 520               | 560   | 447  |
| Other                               |                |     |     |     | 1   |     | 1   | 1   | 1   | 7    | 1                 | 7     | 4    |
| Interviews with external informants |                |     |     |     |     |     |     |     |     |      |                   |       |      |
| Primary                             |                |     |     |     |     |     |     | 8   | 2   | 13   |                   |       |      |
| Archival                            |                | 2   | 2   | 1   | 10  | 17  | 11  | 8   | 4   | 2    |                   |       |      |

*Quotes are attributed in-text with the following codes: Primary interview (PI), Archival interview (AI), Archival presentation (AP), Blog post (BP), Company press release (PR), News article (NA), Other (OT). External informants cited by title.*

**Table 3: Theories at founding**

| <b>Firm</b>      | <b>Initial perception of industry</b>                                                                                                                                                                                                                         | <b>Theory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Illustrative quotes</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Earthtech</b> | <p>Hardware-focused incumbents limited by lack of specialization and coordination</p> <p><i>The kind of people inhabiting [solar] at that time were pioneering engineers who did not have the customer-service component (E1, 2009, AI)</i></p>               | <p><b>Problem to solve:</b> How to facilitate the growth of the solar industry</p> <p><b>Logic:</b> Industries take off when participants are able to specialize and cooperate, as complementary specialization drives costs down</p> <p><b>Implied solution:</b> Innovate in remote sales and design, cultivate a collaborative partner network for finance and installation, allowing each firm to specialize</p>                                                      | <p><i>Much like Dell Computer revolutionized the PC industry and Netflix movie rentals, our goal is to jump start the residential solar industry by making it simple and efficient for everyone with a view of the sun to go solar” (E1, 2008, NA)</i></p> <p><i>Let’s celebrate the mom and pop shops who persist with clunky channels to get PV on peoples’ roofs. Even more key - let’s welcome new players making it easier to bring sunshine online everywhere (E1, 2009, AP)</i></p> <p><i>Installers love to install, but they do not like doing sales. We are taking away the work that they do not like doing (E1, 2009, AI)</i></p>                                                                                                                            |
| <b>Megasolar</b> | <p>Largely backwards incumbents operating below minimum efficient scale</p> <p><i>It’s all mom-and-pop contractors, and they basically suck (Chair, 2007, NA)</i></p> <p><i>There is ample opportunity to establish a nationwide brand (M1, 2006, NA)</i></p> | <p><b>Problem to solve:</b> How to gain market share over rival firms in a fragmented industry</p> <p><b>Logic:</b> Marginal costs fall when firms can leverage economies of scale and scope; lower costs in turn allow firms to capture market share and grow, further reducing costs</p> <p><b>Implied solution:</b> Build an integrated sales, installation, and finance firm by acquiring and integrating rivals, and leveraging greater scale and scope to grow</p> | <p><i>Our vision is to reduce the cost of solar. The way to do this is to become more efficient at installing solar... If you go to one community and do 50 or 100 homes at a time, you have tremendous amount of efficiencies (M1, 2007, NA)</i></p> <p><i>None of [the incumbents] focused on operational efficiency; none of them had any thought of dropping their own costs. We realized, well this is a very fragmented market and we can get into this through bringing economies of scale, and making it a better value offering for the homeowner (M2, 2011, BP)</i></p> <p><i>It [was] clear that acquiring more of the value chain is what’s critical to drive growth, to pump the working capital into more sales and more installs (Exec, 2014, PI)</i></p> |
| <b>Suncorp</b>   | <p>Largely backwards industry built on outdated business models</p> <p><i>There aren’t very many opportunities where you can so easily undercut \$6 billion in annual sales (S2, 2007, NA)</i></p>                                                            | <p><b>Problem to solve:</b> How to introduce novel and profitable business models in a stagnant industry</p> <p><b>Logic:</b> Profits accrue to the firms that introduce innovation in critical bottleneck components</p> <p><b>Implied solution:</b> Enter commercial finance as a B2B provider, while outsourcing less critical activities to partners</p>                                                                                                             | <p><i>I looked out, and it was grey. It kind of hit me right then and there that this is a real issue. This is our earth, and it’s my responsibility as a businessperson to try to solve these problems with a viable business model (S1, 2009, AI)</i></p> <p><i>I had a background in finance as did my partner, so the way we decided to go after it first was, let’s solve that finance, that business model problem (S1, 2015, AI)</i></p>                                                                                                                                                                                                                                                                                                                          |

**Table 4: Selected examples of asymmetric learning from strategic actions**

| Firm             | Example action                                                 | Observed outcome                                                                                                                        | Asymmetric impact on beliefs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                                                                |                                                                                                                                         | ...regarding how to organize                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ...regarding the problem to solve and logic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Earthtech</b> | 2007 Partner with Megasolar and Suncorp for access to finance  | Megasolar unwilling to partner<br>Suncorp only willing to partner under extractive terms and agreement that Earthtech remains exclusive | <b>Revised</b> belief regarding the importance of lease financing; decision to focus on using technology-enabled sales to drive down costs<br><br>“We need to ramp up and innovate, and the Internet is going to be a key channel to getting to customers and making it easy and low cost” (E1, 2008, NA)<br><br>“We were like, this is an entirely commoditizable activity, project finance happens all around us...this is a new asset class and its bumpy right now but over time, it’s going to be just like anything else” (E2, 2014, PI) | Belief that the industry is being held back by lack of cooperation is <b>reaffirmed</b> , as evidenced by Suncorp and Megasolar prioritizing their own competitive advantage over a possible partnership<br><br>“The way [scaling] will happen is not this weird market for hot air traders that just get to create commodities out of something fictional and make a buck. It’s going to be through us driving the demand as we are in California to scale up the industry” (E1, 2008, NA)<br><br>“It’s always been so weirdly competitive” (E1, 2014, PI) |
| <b>Megasolar</b> | 2008 Attempt to develop an in-house tax equity finance product | Effort to recruit banks to provide capital for the lease finance product are unsuccessful, as banks reveal investment size is too small | <b>Revised</b> belief regarding how to get to scale; specifically, realization that a radical new “community-based” sales approach is necessary to reduce costs<br><br>“We will go into a local community and educate the community about the benefits of solar, and try to get the entire community to convert to solar. By doing that, we get efficiencies of scale and we offer a discount, so it’s a win-win” (M1, 2008, AI)                                                                                                               | Belief in importance of scale is <b>reaffirmed</b> , providing explanation for why the effort failed<br><br>“Unless you’re going to do large volume with the bank, they don’t want to bother” (M1, 2010, AI)<br><br>“In 2007, the biggest reason we weren’t able to offer the leasing program was the volume was not interesting to the bank” (M1, 2010, AI)                                                                                                                                                                                                |
| <b>Suncorp</b>   | 2007 Entry into the commercial finance market                  | Suncorp’s initial project bids are unsuccessful, as market is revealed to be more competitive than expected, with lower margins         | <b>Revised</b> belief about which market to enter; specifically, realization that the residential market is more backwards and ripe for innovation<br><br>“We very quickly discovered that everybody was trying to solve the problem of building big commercial solar projects, but nobody was focusing on residential... We felt it would be a better, more defensible business over time” (S1, 2010, AI)                                                                                                                                     | Importance of identifying and innovating in underserved markets is <b>reaffirmed</b> , as failure in commercial finance highlights challenges of being in a competitive market<br><br>“Executing a business is much harder than finance people think. The investment world and the academic world underestimate the advantages and difficulties of executing well” (S2, 2013, NA)                                                                                                                                                                           |

**Table 5: Selected examples of stable beliefs enabling strategic novelty**

| Stable elements of theory                                                                                           | Novel solution considered                                                       | Role of theory                                                                                                                                                     |                                                                                                                                                                                 | Subsequent changes to firm activity system                                                                                               |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                     |                                                                                 | <i>Surfacing a novel opportunity</i>                                                                                                                               | <i>Justifying potential action to self and others</i>                                                                                                                           |                                                                                                                                          |
| <b>Earthtech</b><br><br><i>Facilitate cooperation and specialization to reduce costs and allow industry to grow</i> | 2011 shift to “social sales” model, rather than technology-enabled remote sales | Internal data suggests that referrals are lower cost, but existing sales operation is not well set up to capitalize on this channel                                | Although disruptive to existing sales process, mobilizing existing customers will allow superior growth; <i>“The idea is solar as a social network”</i>                         | <b>Degree of change:</b><br>+++++<br>Product feature shift<br>Changes to sales team compensation<br>New advertising campaign<br>CMO hire |
|                                                                                                                     | 2014 Begin sourcing hardware for partners, having earlier exited installation   | Observation of Megasolar’s lower costs conflicts with founders’ belief in specialization; prompts search for ways to help installer partners                       | Investment in installation framed as a logical way to help partners; <i>“Rather than let the little guys do the buying, we do it, negotiate better with deals and credit”</i>   | <b>Degree of change:</b><br>++<br>2 new “labs” opened<br>New hardware sourcing partnerships<br>Large financial investment                |
| <b>Megasolar</b><br><br><i>Leverage economies of scale and scope to reduce costs</i>                                | 2010 Enter energy efficiency (EE) services                                      | Observation of growth of EE market, which is viewed as an opportunity to “offer more services and more value”, thus expanding scope and thus increasing efficiency | Upfront investment and costly acquisition are framed as a way to drive future growth and reduce costs; <i>“We’re simply focused on bringing economies of scale”</i>             | <b>Degree of change:</b><br>++<br>4 partnerships<br>1 acquisition<br>New line of business                                                |
|                                                                                                                     | 2013 Enter racking (hardware component)                                         | Novel hardware innovation seen as defensible and a way to improve installation efficiency, entry into hardware had never been considered                           | Up front disruption to installation routines will result in future improvement in installation efficiency; <i>“This will double throughput [and cut install time in half]”</i>  | <b>Degree of change:</b><br>+++++<br>\$158M acquisition<br>Transition into hardware manufacturing<br>Shift in installation routines      |
| <b>Suncorp</b><br><br><i>Innovate in underserved markets to create value and bargaining power</i>                   | 2014 Enter installation                                                         | Shifts in the industry have made installation critical: <i>“There is an opportunity now to have scale advantages in...installation”</i>                            | Entering installation framed as a way to generate learning and improve bargaining power: <i>“[This] is merely the next step in the evolution of a rapidly growing business”</i> | <b>Degree of change:</b><br>+++++<br>~\$120M acquisition<br>Transition into installation<br>Integrate 800+ new employees                 |
|                                                                                                                     | 2016 Enter energy storage                                                       | Observation of energy storage as a new, underserved market; <i>“An enormous transformation is underway”</i>                                                        | Expanding services provided framed as a way to “increase customer lifetime value” by cross-selling products and services                                                        | <b>Degree of change:</b><br>++<br>2 partnerships<br>New product release<br>Changes to sales and installation routines                    |

**Figure 1: Key strategic actions and select performance milestones by firm**

|                  | '07                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | '08 | '09 | '10 | '11 | '12 | '13 | '14 | '15 | '16 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| <b>Earthtech</b> | <p>○ Launch</p> <p>○ Expand to new states</p> <p>○ Expand internationally (4x)</p> <p>○ <b>Build remote design technology</b></p> <p>○ Partner with major nonprofit</p> <p>○ Enter commercial market</p> <p>○ Partner with major nonprofit</p> <p>○ Partner with major entertainment provider</p> <p>Δ Drop existing installer partners, form new tie portfolio</p> <p>Δ <b>Launch novel “solar incubator”</b></p> <p>Δ Forgo finance</p> <p>Δ Internalize finance</p> <p>Δ Exit and outsource finance</p> <p>Δ <b>Internalize installation</b></p> <p>Δ Adopt radical new “social sales” model</p> <p>Δ Exit and outsource installation</p> <hr/> <p>✓ Top-ten seller of small (&lt;10kW) systems</p> <p>✓ \$30M revenue, <b>30% lower costs vs industry average</b></p> <p>✓ Fastest growing US solar company</p> <p>✓ <b>Top ten largest US solar company</b></p>                                                                                                                                                                                                                         |     |     |     |     |     |     |     |     |     |
| <b>Megasolar</b> | <p>○ Launch</p> <p>○ Expand to new states</p> <p>○ <b>Acquire installers to further expand</b></p> <p>○ Expand internationally</p> <p>○ <b>Acquire and roll-up incumbent installers</b></p> <p>○ <b>Invest in operations software and streamlining sales</b></p> <p>Δ Launch utility services division</p> <p>Δ Enter energy efficiency services</p> <p>Δ Exit energy efficiency services</p> <p>○ Partner with major retailer</p> <p>Δ Launch without finance</p> <p>○ Launch finance after delay, decline to offer to competitors</p> <p>○ Launch novel consumer loan product</p> <p>Δ Debut novel “community sales” model</p> <p>Δ Undertake major internal incentive redesign</p> <p>Δ Internalize racking</p> <p>Δ Enter electric vehicle charging infrastructure</p> <p>Δ <b>Entry into solar panel manufacturing</b></p> <hr/> <p>✓ \$23M revenue</p> <p>✓ 10,000 cumulative customers</p> <p>✓ \$400M revenue</p> <p>✓ <b>Raise \$10M seed round</b></p> <p>✓ 34% residential market share</p> <p>✓ \$1B cumulative project financing</p> <p>✓ \$8B cumulative project financing</p> |     |     |     |     |     |     |     |     |     |
| <b>Suncorp</b>   | <p>○ Launch</p> <p>○ <b>Partner with 10+ top installers</b> to expand</p> <p>Δ Internalize installation and racking</p> <p>Δ Exit commercial, <b>develop lease</b> to enter residential market</p> <p>○ Partner with home builder</p> <p>○ Partner with retailer</p> <p>○ <b>Secured investment from major banks for finance</b></p> <p>Δ Enter energy storage</p> <p>Δ Enter smart home services</p> <p>Δ <b>Launch internal sales operation</b></p> <p>○ Major expansion of internal sales</p> <p>○ Launch loan product</p> <p>○ Expand partner network (30 installers)</p> <hr/> <p>✓ <b>\$12M Series A</b></p> <p>✓ &gt;\$1B in cumulative project financing</p> <p>✓ &gt;\$2B in cumulative project financing</p> <p>✓ <b>25,000 customers and growing fast</b></p> <p>✓ \$55M revenue</p> <p>✓ Sustained profitability</p>                                                                                                                                                                                                                                                             |     |     |     |     |     |     |     |     |     |

Key: ✓ reflect performance milestones. Δ represent major changes to ventures' existing activities (exhaustive list). ○ represent important resource commitments that elaborate on existing activities (including only those cited as particularly important by informants). **Activities we discuss in the narrative are in bold.**



**Figure 2: Propositions for a model of theory enactment**

